gGA Dimer Scans: U/t Dependence at $\Delta V = 0$ and $\Delta V = 2$

(Ghost Multiplicities $N_g = 2$ and $N_g = 4$, Direct-Minimization Solver)

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1 Total energy and error vs. exact: $N_g = 2$ and $N_g = 4$

Figures 1 and 2 show the variational energy E_{var}/t of the $N_g = 2$ and $N_g = 4$ solutions together with the exact (ED) energy, and the error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ on a logarithmic scale, for $\Delta V = 0$ and $\Delta V = 2$ respectively. For $\Delta V = 2$ the plotted gGA energies are the cross-checked ("confirmed") solutions. The two N_g curves are indistinguishable on this scale. At $\Delta V = 0$, $\max |E(N_g{=}4) - E(N_g{=}2)| = 3 \times 10^{-8}$. At $\Delta V = 2$ the two agree to better than 10^{-6} for $U/t \lesssim 6.2$, while for $U/t \gtrsim 6.3$ the $N_g = 4$ energy is lower by up to 1.3×10^{-4} (at $U/t = 8.4$); this is still negligible against the error to the exact energy, ≈ 0.3–0.4 there (see Sec. 3).

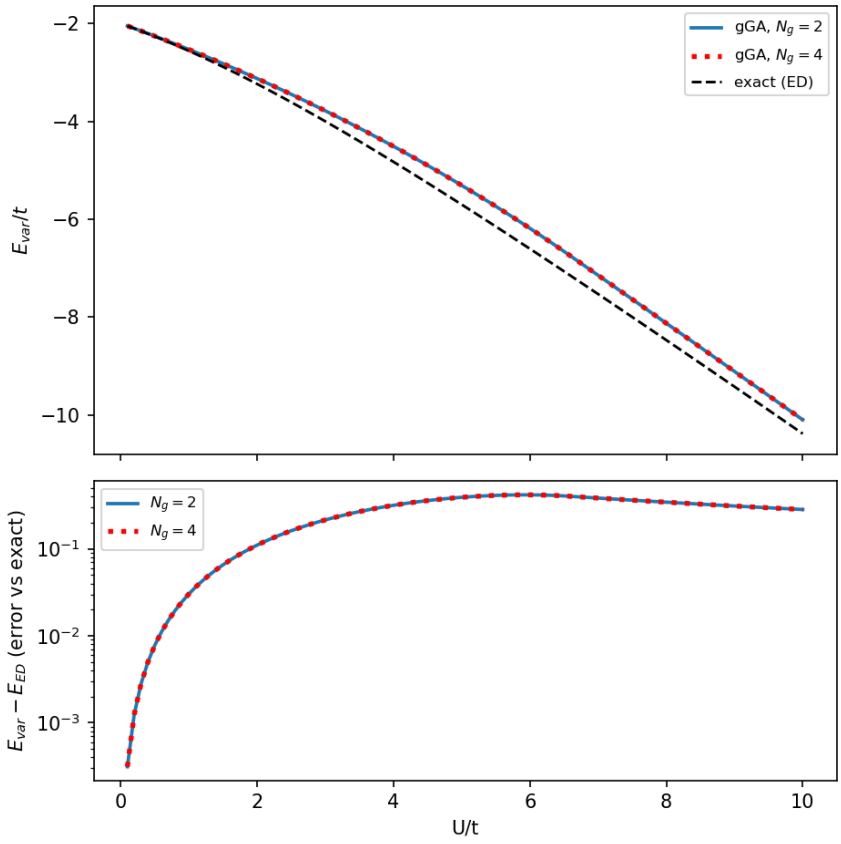


Figure 1: $\Delta V = 0$: E_{var}/t (top) and error $E_{\text{var}} - E_{\text{ED}}$ (bottom) vs. U/t for $N_g = 2$ and $N_g = 4$.

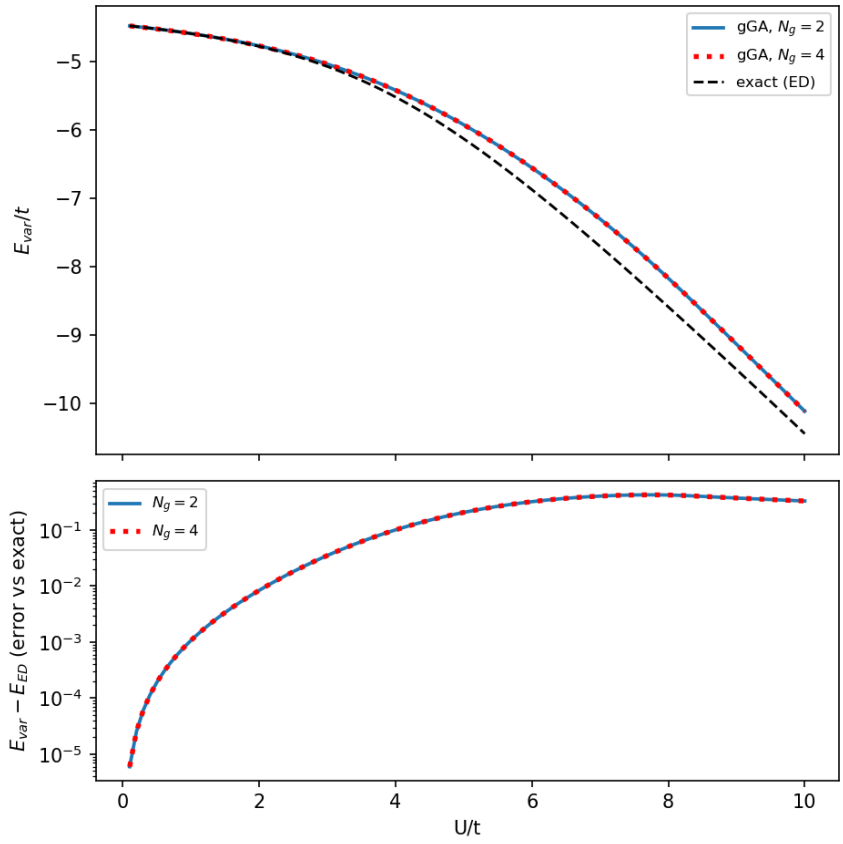
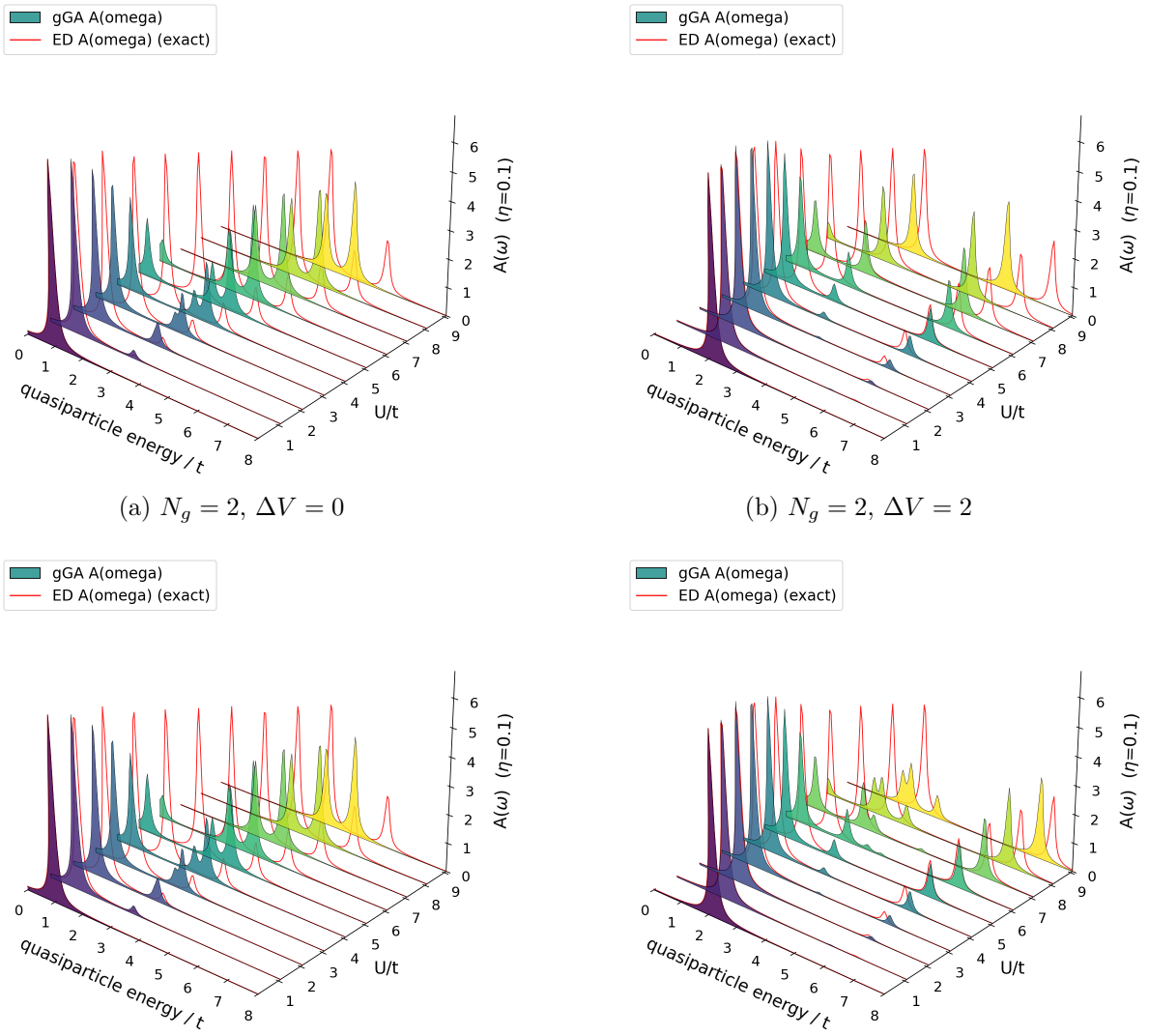


Figure 2: $\Delta V = 2$: E_{var}/t (top) and error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ (bottom) vs. U/t for $N_g = 2$ and $N_g = 4$.

2 Spectra and spectral-function movies: $N_g = 2$ and $N_g = 4$

2.1 3D view

Figure 3 shows, for each $(N_g, \Delta V)$ combination, the gGA spectral function $A(\omega)$ (filled ridgelines, built from the quasiparticle poles and weights of the converged solution, broadening $\eta = 0.1$) as a function of U/t, together with the exact (ED) spectral function (red lines). Rows are $N_g = 2$ and $N_g = 4$, columns are $\Delta V = 0$ and $\Delta V = 2$. The $\Delta V = 2$ panels use the cross-checked solutions.



(a) $N_g = 2$, $\Delta V = 0$

(b) $N_g = 2$, $\Delta V = 2$

(c) $N_g = 4$, $\Delta V = 0$

(d) $N_g = 4$, $\Delta V = 2$

Figure 3: gGA spectral function $A(\omega)$ (filled) vs. ED (red) as a function of U/t for $N_g = 2$ (top) and $N_g = 4$ (bottom), at $\Delta V = 0$ (left) and $\Delta V = 2$ (right). Only $\omega \geq 0$ is shown: $A(\omega) = A(-\omega)$ in both gGA and ED.

Analysis. The number of quasiparticle branches is $2(1 + N_g)$: six for $N_g = 2$ and ten for $N_g = 4$ (not a doubling). In all four panels the gGA peaks coincide with the ED ones at small U/t and move away from them as U/t grows, consistent with the error in the total energy growing with U/t (Sec. 1). The $N_g = 4$ ridgelines are visually indistinguishable from the $N_g = 2$ ones at the same ΔV: the extra branches carry no visible spectral weight, in line with the energies, so the gGA peaks do not follow the ED peaks more closely over the scanned range; the additional branches are the ones discussed as "sleeping" ghosts in Sec. 3. The effect of ΔV is visible in the same way for both N_g: at $\Delta V = 2$ the branches are well separated at small U/t and bunch together as U/t increases, with a rearrangement between $U/t = 0.6$ and 0.7 (Sec. 3).

2.2 Spectral function movies

- $N_g = 2$, $\Delta V = 0$: spectral-function movie
- $N_g = 2$, $\Delta V = 2$: spectral-function movie
- $N_g = 4$, $\Delta V = 0$: spectral-function movie
- $N_g = 4$, $\Delta V = 2$: spectral-function movie

(Links open the movie with the PDF viewer's associated application, if the viewer supports `run:` links.)

Quasiparticle-weight movies. The same scans are shown level by level: for each U/t the eigenvalues ϵ_n^{qp} of H_{qp} with the weight carried by each level (summed over both sites and spins, so the total weight is that of $A(\omega)$), and the ED poles as red open circles (linear and logarithmic scales; script `make_qp_weights_video.py`). A level with weight $\geq 10^{-3}$ is a peak (levels without weight are light gray). At $N_g = 2$ a peak keeps its color along U/t (consecutive U/t points are matched by closeness in energy and weight), and the colors **blue**, **orange**, **green** follow the order of increasing energy at the U/t where all the peaks coexist. At $N_g = 4$ a peak that corresponds to an $N_g = 2$ peak (matched at the same U/t) has its color, so the same physical peak has the same color at $N_g = 2$ and 4; the extra $N_g = 4$ peaks are **purple** and **brown**, in order of increasing energy.

- $N_g = 2$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 2$, $\Delta V = 2$: quasiparticle-weight movie
- $N_g = 4$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 4$, $\Delta V = 2$: quasiparticle-weight movie

Weight transfer. Positive-energy side. Each peak is written as (weight, energy) in the color of its peak in the movies (weights summed over sites and spins, total 2.0; ED poles in red). At $\Delta V = 0$ the peaks are the blue one, which carries almost all the weight at small U/t, and the orange and green ones, which are born at $U/t \simeq 0.5$ near $\epsilon = 3$. As U/t grows the blue peak loses weight to the orange and green ones while its energy goes to zero, and both vanish at a critical U_c/t ($N_g = 2$ and 4 agree; ED has no such event; the values come from a linear extrapolation of the last grid points); the orange and green peaks continue, with the same colors, above U_c.

- $\Delta V = 0$, $U/t \lesssim 1$ ($U/t = 1$): (1.92, 0.97) carries $\geq 96\%$; (0.02, 2.99) and (0.06, 3.02) hold the rest. ED: (1.97, 1.06), (0.03, 3.06).
- $\Delta V = 0$, $1 \lesssim U/t < U_c/t \simeq 6.56$: the blue peak loses its weight and the other two gain it. $U/t = 4$: (1.08, 0.57), (0.27, 2.83), (0.66, 3.26); $U/t = 6$: (0.27, 0.14), (0.66, 2.56), (1.05, 3.41); $U/t = 6.5$: (0.03, 0.02), (0.80, 2.46), (1.13, 3.43). ED at $U/t = 4$: (1.71, 1.83), (0.29, 3.83).
- $\Delta V = 0$, $U/t > U_c/t$: only the orange and green peaks. $U/t = 7$: (0.83, 2.69), (1.13, 3.67); $U/t = 10$: (0.89, 4.29), (1.10, 5.29). ED at $U/t = 10$: (1.37, 4.39), (0.63, 6.39).

At $\Delta V = 2$ the blue peak is again the main one; the orange peak is the middle one (born at $U/t \simeq 1.5$–3) and the green one the upper one (born at $U/t \simeq 1$).

- $\Delta V = 2$, $U/t \lesssim 3$ ($U/t = 3$): (1.89, 1.25), (0.015, 3.59), (0.10, 5.99); ED: (1.89, 1.34), (0.11, 5.81).
- $\Delta V = 2, 3 \lesssim U/t < U_c/t \simeq 8.3$: the blue peak loses weight and the other two gain it. $U/t = 5$: (1.46, 0.81), (0.18, 2.57), (0.36, 5.90); $U/t = 6$: (1.10, 0.59), (0.37, 2.33), (0.53, 5.83); $U/t = 8$: (0.18, 0.09), (0.84, 2.08), (0.95, 5.28). The blue peak has weight 0.02 at $U/t = 8.3$ and 10^{-3} at 8.5 (it stays at $\epsilon \lesssim 10^{-3}$).

- $\Delta V = 2$, $U/t > U_c/t$ ($N_g = 2$): only the orange and green peaks. $U/t = 9$: (0.94, 2.38), (1.02, 5.66); $U/t = 10$: (0.96, 2.88), (1.02, 6.36). ED at $U/t = 10$: (1.22, 3.21), (0.78, 7.68).

Above U_c the two weights approach each other but are not equal up to $U/t = 10$; gGA is closer to equal sharing than ED, i.e. it puts too much weight on the upper level. For $N_g = 4$ there are four more levels, near $\epsilon = 0$ and without weight at small U/t (colors as in the movies). At $\Delta V = 0$ the peaks are those of $N_g = 2$ (same colors and values, differences $\leq 2 \times 10^{-3}$) for all U/t, including U_c/t: the extra levels never carry weight. At $\Delta V = 2$ the regions are:

- $\Delta V = 2$, $U/t \lesssim 5.3$: as $N_g = 2$ (the smallest peaks differ by $\lesssim 6 \times 10^{-3}$ in weight).
- $\Delta V = 2$, $5.4 \lesssim U/t < 7.5$: two extra peaks appear, purple and brown, and come and go with U/t. $U/t = 6$: (1.09, 0.59), (0.34, 2.27), (0.05, 3.01), (0.03, 3.26), (0.50, 6.15), against (1.10, 0.59), (0.37, 2.33), (0.53, 5.83) at $N_g = 2$; $U/t = 7$ (brown below the weight threshold): (0.66, 0.35), (0.53, 2.15), (0.16, 2.56), (0.66, 6.31), against (0.66, 0.35), (0.60, 2.18), (0.73, 5.65). The purple peak takes its weight from the orange and green ones, and the blue one is not changed.
- $\Delta V = 2$, $U/t = 7.5 \to 7.6$: the solution changes discontinuously (five peaks to four, the energy of the green peak jumps from 6.38 to 5.48).
- $\Delta V = 2$, $7.6 \lesssim U/t < U_c/t \simeq 8.3$: five peaks; $U/t = 8$: (0.17, 0.09), (0.38, 1.79), (0.43, 2.08), (0.18, 3.06), (0.85, 6.44). The blue peak vanishes at about U_c (0.014 at $U/t = 8.3$), as at $N_g = 2$.
- $\Delta V = 2$, $U/t > U_c/t$: four peaks. $U/t = 9$: (0.35, 2.00), (0.46, 2.29), (0.27, 3.22), (0.92, 6.82); $U/t = 10$: (0.34, 2.53), (0.58, 2.85), (0.15, 4.14), (0.93, 7.32), against (0.96, 2.88), (1.02, 6.36) at $N_g = 2$ and (1.22, 3.21), (0.78, 7.68) in ED. The weight of the orange peak of $N_g = 2$ (0.96) is shared by the orange peak and the two extra ones (1.07 in total), and the green peak moves toward ED (7.32 against 6.36 at $N_g = 2$; ED 7.68) and loses weight (0.93, ED 0.78).

The $N_g = 4$ departure from $N_g = 2$ starts at $U/t \simeq 5.4$, consistent with the $N_g = 4$ energy departing from the $N_g = 2$ one for $U/t \gtrsim 6.2$ (Sec. 1).

2.3 Quasiparticle orbitals and density: the gap closing at U_c

Two more movies per case show the eigenorbitals of H_{qp} and the quasiparticle density as U/t grows ($N_g = 2, 4$; $\Delta V = 0, 2$; script `make_qp_orbital_videos.py`). They are drawn on the skeleton of the dimer: the two large circles joined left to right are the two states (the direction of R_s, the only qp orbital that hops to the other site), and the small outer circles are their ghosts (the on-site eigenvectors orthogonal to it). Occupation as in the code (`gga_dimer.qp_state`): the lowest $N/2$ eigenvectors returned by `eigh` are filled, without fractional occupation. In the orbital movies the occupied orbitals are in the first row and the empty ones in the second, each row in order of increasing energy; circle area is $|\text{amplitude}|^2$ and the color is the sign (signs are aligned with the previous U/t by maximum overlap). In the density movies the colors give the qp density per orbital (the code's ρ), next to the gaps, bond order, Z and double occupancy (or state density at $\Delta V = 2$) as functions of U/t; the dashed line is U_c/t.

- $N_g = 2$, $\Delta V = 0$: orbitals, density
- $N_g = 2$, $\Delta V = 2$: orbitals, density
- $N_g = 4$, $\Delta V = 0$: orbitals, density
- $N_g = 4$, $\Delta V = 2$: orbitals, density

What can be seen at the critical U_c/t of the weight transfer (6.56 for $\Delta V = 0$, $\simeq 8.3$ for $\Delta V = 2$):

- *Spectral gap* (first panel of the density movies, lowest positive minus highest negative energy among the levels with weight $\geq 10^{-3}$): it decreases smoothly with U/t (2.0 at $U/t = 0.1$, 1.14 at 4, 0.27 at 6 for $\Delta V = 0$; 4.39, 2.0 (at $U/t = 4$) and 1.19 (at 6) for $\Delta V = 2$, $N_g = 2$)

and closes at U_c; the next grid point has a large gap ($0.03 \to 4.94$ between $U/t = 6.5$ and 6.6 at $\Delta V = 0$; $0.001 \to 4.36$ between 8.5 and 8.6 at $\Delta V = 2$, $N_g = 2$; $0.014 \to 3.30$ between 8.3 and 8.4 at $N_g = 4$), and it grows afterwards (at $U/t = 10$: 8.59 and 5.76). The qp gap (HOMO–LUMO) is zero from U_c on; before U_c it follows the spectral gap, except at small U/t, where weightless ghost levels lie inside the gap.

- *Orbitals*: below U_c the HOMO and LUMO have amplitude on the two states and carry $A(\omega)$ weight; from U_c on, the orbitals at $\epsilon = 0$ have no amplitude on the states and zero weight, and sit on ghost circles only (two of them at $N_g = 2$, six at $N_g = 4$). Which of these degenerate orbitals are occupied is the choice of `eigh`, so the occupied ghost orbitals change from frame to frame above U_c.
- *Bond order and double occupancy* (center–center density matrix element, per spin): both change slope at U_c (at $\Delta V = 0$: bond order 0.17, 0.09, 0.08 and double occupancy 0.040, 0.016, 0.013 at $U/t = 6$, 6.5, 6.6). At $\Delta V = 2$ the bond order has a maximum (0.40) at $U/t \simeq 4.3$ and falls to 0.08 at U_c.
- *State density at* $\Delta V = 2$: the densities of the two states are polarized at small U/t (0.945 and 0.055 at $U/t = 0.1$, 0.63 and 0.37 at 6) and become equal at U_c (0.50 and 0.50, $N_g = 2$), staying equal above.
- Z stays ≥ 0.975 for all U/t (no collapse of the renormalization), and at $\Delta V = 0$ the state densities are 0.50 at all U/t; the colors of the ghost circles are not a signature of the transition.

This is a gap closing and re-opening of the gGA solution. It is not present in the exact dimer, whose spectral gap varies smoothly with U/t (Sec. 2).

2.4 Ionic and covalent weights

As U/t grows (the analogue of stretching H_2) the weight of the ionic configurations (one site doubly occupied, H^-H^+) is expected to go to the covalent ones (one electron on each site). The ionic weight is $P_{\text{ion}} = \langle n_{0\uparrow} n_{0\downarrow} \rangle + \langle n_{1\uparrow} n_{1\downarrow} \rangle$ and the covalent weight is $1 - P_{\text{ion}}$. In gGA the double occupancies and the site occupations come from the impurity ground state of each fragment (`gga_dimer.impurity_solve`) at the converged (R, λ), as in one iteration of the loop (at $\Delta V = 0$ they reproduce the `docc` of the solver output); ED uses the exact ground state ($\Delta V_{\text{ED}} = 2\,\Delta V_{\text{gGA}}$). $N_g = 2$ and 4 agree to 3–4 digits in P_{ion} (checked at $\Delta V = 2$).

	$\Delta V = 0$: P_{ion}		$\Delta V = 2$: P_{ion} and electrons on the favored site			
U/t	ED	gGA	P_{ion} ED	P_{ion} gGA	n ED	n gGA
0.1	0.488	0.494	0.896	0.896	1.890	1.890
2	0.276	0.374	0.767	0.781	1.755	1.764
4	0.146	0.239	0.469	0.557	1.446	1.517
6	0.084	0.079	0.205	0.310	1.178	1.261
7	0.066	0.022	0.138	0.193	1.113	1.154
8.5	0.048	0.015	0.083	0.029	1.061	1.021
10	0.036	0.010	0.055	0.017	1.036	1.011

Table 1: Ionic weight (ED and gGA, $N_g = 2$) and, at $\Delta V = 2$, electrons on the favored site ($n = 1$ is purely covalent).

At $\Delta V = 0$ there is no charge transfer ($n = 1$ on each site) and the ionic weight falls from 0.5, i.e. the weight moves from ionic to covalent, in ED smoothly (0.036 at $U/t = 10$). gGA does the same but is too ionic at intermediate U/t and then drops across $U_c/t \simeq 6.56$ ($0.079 \to 0.022$

between $U/t = 6$ and 7) to an almost purely covalent state (0.010 at $U/t = 10$). At $\Delta V = 2$ the state is polarized (H^-H^+-like, 1.89 electrons on the favored site) and becomes covalent as U/t grows; gGA is more ionic than ED up to $U/t \simeq 7$ and less ionic after $U_c/t \simeq 8.3$ ($n = 1.021$ against 1.061 at $U/t = 8.5$). The density of the two states in the density movies matches the physical occupation per spin at the two points checked (0.945 at $U/t = 0.1$ and 0.630 at 6 for $\Delta V = 2$).

Link with the spectral weights. The hole peaks of $A(\omega)$ are photoemission lines: an electron is removed from the dimer and the cation is left in its bonding (lowest) or antibonding one-electron orbital; the inner pole is the bonding final state and the outer pole the antibonding one. The intensity of each line is the overlap between that final state and the state obtained by removing an electron from the ground state. At $\Delta V = 0$ write the ground state as $\cos\theta\,|\mathrm{cov}\rangle + \sin\theta\,|\mathrm{ion}\rangle$ ($P_{\mathrm{ion}} = \sin^2\theta$). Removing an electron from the covalent part leaves the other electron on the *other* site, with amplitude $\propto \cos\theta$; removing it from the ionic part leaves the remaining electron on the *same* site, with amplitude $\propto \sin\theta$. The two contributions interfere: they add in the bonding orbital and subtract in the antibonding one, so that

$$w_{\mathrm{inner}} = (\cos\theta+\sin\theta)^2 = 1+2\sqrt{P_{\mathrm{ion}}(1-P_{\mathrm{ion}})}, \qquad w_{\mathrm{outer}} = (\cos\theta-\sin\theta)^2 = 1-2\sqrt{P_{\mathrm{ion}}(1-P_{\mathrm{ion}})},$$

which are the exact weights $1 \pm 4t/\Omega$, $\Omega = \sqrt{U^2+16t^2}$ (checked numerically). For example, at $U/t = 4$, $\cos\theta = 0.924$ and $\sin\theta = 0.383$ ($P_{\mathrm{ion}} = 0.146$) give 1.707 and 0.293. The imbalance between the two lines is thus the product of the covalent and ionic amplitudes: it is maximal at $U = 0$ ($P_{\mathrm{ion}} = \frac{1}{2}$, weights 2 and 0, the one-electron picture) and it disappears when the state is purely covalent (both weights equal to 1, the two Hubbard-band-like lines). Transferring weight from the inner to the outer pole as U/t grows is the same process as transferring weight from ionic to covalent. At $\Delta V = 2$ the same interference holds but the ionic amplitudes of the two sites differ, so the weights depend on three amplitudes (covalent, ionic on each site).

gGA does not compute the weights this way: they come from R and the quasiparticle spectrum, not from the ionic and covalent amplitudes of the ground state, so the link is only partly kept. Inserting gGA's own composition in the exact relation gives the inner weight that the gGA state would have; it is compared below with the weight of the lowest-energy peak of the gGA spectrum (nearest to $\epsilon = 0$; $N_g = 2$):

	$\Delta V = 0$: inner weight			$\Delta V = 2$: inner weight		
U/t	ED	from gGA P_{ion}	gGA peak	ED	from gGA composition	gGA peak
2	1.894	1.968	1.717	1.965	1.971	1.963
4	1.707	1.852	1.076	1.760	1.830	1.729
6	1.555	1.541	0.266	1.472	1.596	1.095
8	1.447	1.256	—	1.306	1.261	0.177
10	1.371	1.203	—	1.222	1.117	—

Table 2: Weight of the inner pole: exact, predicted from the ionic/covalent composition of the gGA state, and the weight of the gGA peak nearest to $\epsilon = 0$ ($N_g = 2$; —: that peak has vanished above U_c).

The qp weight follows the exact value for $U/t \lesssim 4$ at $\Delta V = 2$, but at $\Delta V = 0$ the lowest peak is already too light at $U/t = 2$ and loses its weight to the other two peaks as it goes to $\epsilon = 0$ at U_c. Above U_c the two remaining peaks are more equal than the exact ones (0.89 and 1.10 at $\Delta V = 0$, 0.96 and 1.02 at $\Delta V = 2$, against 1.37 and 0.63, 1.22 and 0.78 at $U/t = 10$), consistent with gGA's

ground state being too covalent at large U/t. The total weight of the positive-energy peaks stays close to 2.

3 Discussion: $\Delta V = 0$ vs. $\Delta V = 2$

Weak coupling: ΔV dominates. At small U/t the two scans differ substantially. At $U/t = 0.1$, $E_{\mathrm{var}}/t = -2.0503$ for $\Delta V = 0$ but -4.4823 for $\Delta V = 2$ (both N_g). This is the expected $U \to 0$ limit: with $\Delta V = 0$ the dimer's noninteracting energy is $-2t = -2$ plus a small $O(U)$ correction, while at $\Delta V = 2$ the site-splitting opens a one-body level gap and the $U \to 0$ energy approaches the closed form $E = -2\sqrt{t^2 + \Delta V^2} = -2\sqrt{5} \approx -4.472$. The gap between the two curves is largest here, $\Delta E \approx 2.43$, because ΔV acts as an unscreened one-body potential that the (still weak) correlations barely renormalize.

Strong coupling: the curves converge. As U/t grows the two energies draw close together: at $U/t = 10$, $E_{\mathrm{var}}/t = -10.101$ ($\Delta V = 0$) vs. -10.120 ($\Delta V = 2$), a difference of only ≈ 0.02 – two orders of magnitude smaller than at weak coupling. In the Mott-like regime the total energy is governed almost entirely by U (both sites approach single occupancy), and the fixed $\Delta V = 2$ site-splitting becomes a small perturbation on top of the dominant Hubbard physics. The quasiparticle spectra (3D plots, Fig. 3) show the same trend: the $\Delta V = 2$ branches, well separated at small U, bunch together with the $\Delta V = 0$-like structure as U/t increases.

Level crossing at $\Delta V = 2$. The $\Delta V = 2$ quasiparticle energies show a sharp rearrangement between $U/t = 0.6$ and $U/t = 0.7$ (e.g. $N_g = 2$: $\epsilon^{qp} \approx -2.00 \to -3.12$), visible as a kink in the 3D plots; this is the level crossing where the ΔV-induced one-body gap stops being the dominant scale and the Hubbard repulsion takes over.

Ghost multiplicity ($N_g = 2$ vs. $N_g = 4$). At both $\Delta V = 0$ and $\Delta V = 2$, doubling the ghost multiplicity from $N_g = 2$ to $N_g = 4$ leaves the lowest variational energy numerically unchanged at $\Delta V = 0$ over the whole scanned range, and at $\Delta V = 2$ for $U/t \lesssim 6.2$ (for larger U/t the $N_g = 4$ energy is lower by at most 1.3×10^{-4}, Sec. 1): at $\Delta V = 0$, $E(N_g{=}4) - E(N_g{=}2) = 0$ to the printed precision at essentially every U/t (largest deviation $\sim 3 \times 10^{-8}$, at the level of solver tolerance), and the confirmed energies show the same agreement at $\Delta V = 2$ (e.g. $U/t = 1.0$: -4.593786 vs. -4.593786). The extra ghost degrees of freedom introduced at $N_g = 4$ therefore stay essentially energetically inert for this dimer at both values of ΔV in this U range – consistent with the known $N_g \geq 2$ "sleeping ghost" caveat: the additional branches exist in the quasiparticle spectrum (compare Fig. 3, which shows ten instead of six ϵ_n^{qp} branches at $N_g = 4$) but do not lower the ground-state energy below what $N_g = 2$ already finds.